

Sprint #0 Report

Instructions

Objectives

- Make decisions on the Solitaire software development project.
- Learn unit testing and GUI programming in the language of your choice.

Deliverables and Grading Policy

Read the "CS 449 Homework Overview" document **carefully** and make the key decisions for the software development. Use the following template to complete your report.

1. Key Decisions of the Solitaire Project (2 points)

Decision Element	Choice / Justification
Object-oriented programming language (if not using OOP, then justify the use of the language)	C++
GUI library (strongly encouraged)	qt
IDE (Integrated Development Environment)	Qt creator i only just now got it working and spent the last week trying to get qt to work in visual studio
xUnit framework (e.g., JUnit for Java)	GoogleTest
Programming style guide (must read it carefully)	Google c++
Project hosting site	Github.com
Other decisions if applicable	

Sample programming style guides:

- Google Java Style Guide: <https://google.github.io/styleguide/javaguide.html>
- Google C++ Style Guide: <https://google.github.io/styleguide/cppguide.html>
- Google Python Style Guide: <https://google.github.io/styleguide/pyguide.html>

2. Unit testing (4 points)

Find a tutorial on the unit test framework you have chosen and write at least two xUnit tests of a program you have written or found elsewhere. Attach here (1) the screenshot of your program execution and (2) the source code of your program.

For example, JUnit5 user guide: <https://docs.junit.org/5.10.3/user-guide/index.html>

<https://github.com/BusinessClown/marble-solitaire> my code

Please email if the link is not working it is set to public

```
Marble_Solitaire X MarbleSolitaire X tic_tac_toe_game_test X
14:25:22: Starting C:\Users\insan\Downloads\style guide met and google
test\marble_solitaire\build\Desktop_Qt_6_11_2_MinGW_64_bit_Debug\tests\tic_tac_toe_game_test.exe...
Running main() from C:\Users\insan\Downloads\style guide met and google
test\marble_solitaire\build\Desktop_Qt_6_11_2_MinGW_64_bit_Debug\_deps\googletest-
src\googletest\src\gtest_main.cc
[====] Running 4 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 4 tests from TicTacToeGameTest
[ RUN ] TicTacToeGameTest.ResetClearsBoardAndSetsFirstPlayer
[ OK ] TicTacToeGameTest.ResetClearsBoardAndSetsFirstPlayer (0 ms)
[ RUN ] TicTacToeGameTest.RejectsOccupiedAndOffBoardMoves
[ OK ] TicTacToeGameTest.RejectsOccupiedAndOffBoardMoves (0 ms)
[ RUN ] TicTacToeGameTest.DetectsDiagonalWinAndRecordsWinningLine
[ OK ] TicTacToeGameTest.DetectsDiagonalWinAndRecordsWinningLine (0 ms)
[ RUN ] TicTacToeGameTest.FullBoardWithoutWinnerIsADraw
[ OK ] TicTacToeGameTest.FullBoardWithoutWinnerIsADraw (0 ms)
[-----] 4 tests from TicTacToeGameTest (0 ms total)

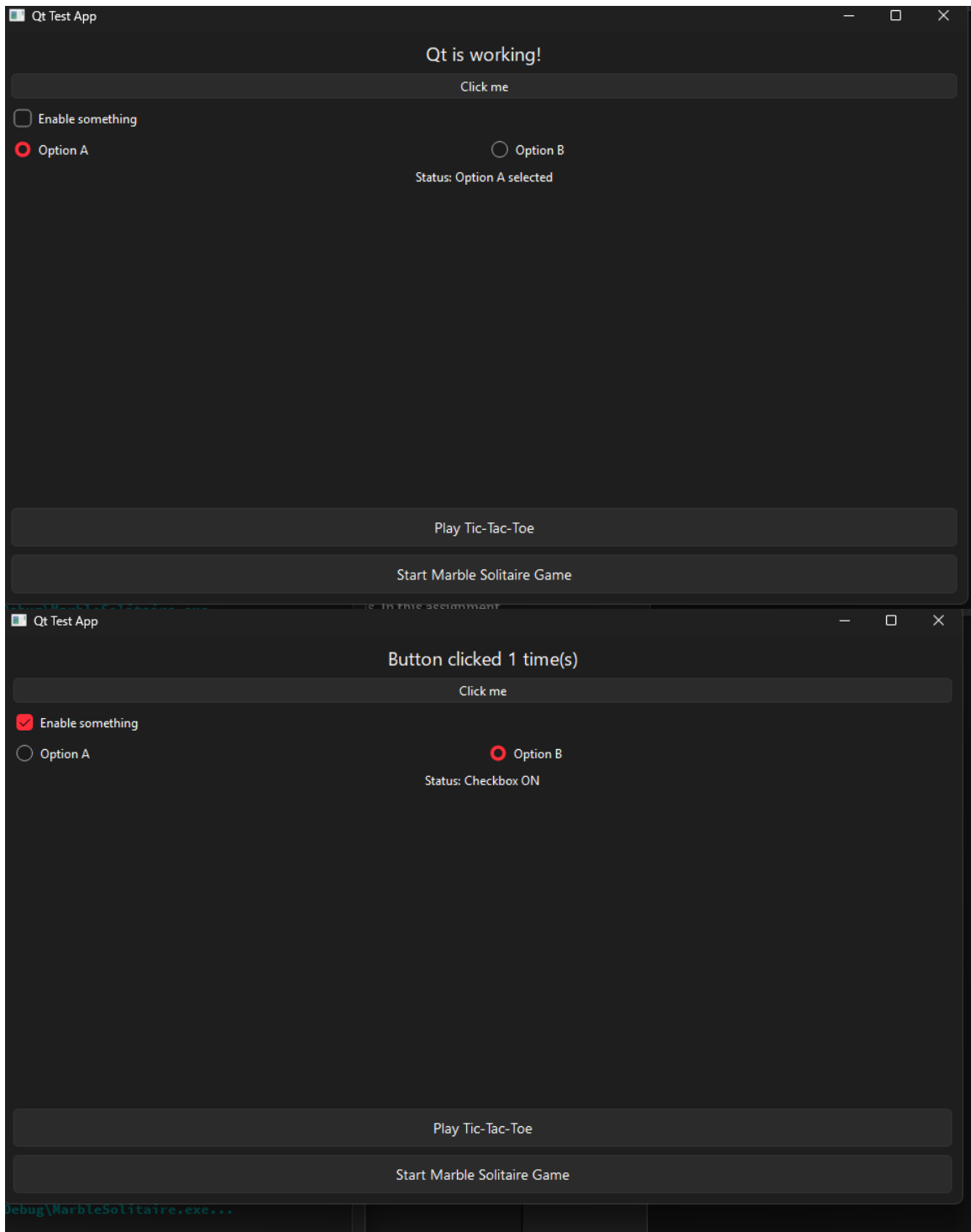
[-----] Global test environment tear-down
[====] 4 tests from 1 test suite ran. (0 ms total)
[ PASSED ] 4 tests.
14:25:22: The command "C:\Users\insan\Downloads\style guide met and google
test\marble_solitaire\build\Desktop_Qt_6_11_2_MinGW_64_bit_Debug\tests\tic_tac_toe_game_test.exe"
finished successfully.
```

3. GUI programming (4 points)

Write a GUI program in the language you have chosen for your Solitaire project. The GUI of your program must include text, lines, a check box, and radio buttons. In this assignment, any GUI program of your own work is acceptable.

Attach here (1) the screenshot of your program execution and (2) the source code of your program.

<https://github.com/BusinessClown/marble-solitaire> my code



Who goes first?

X goes first

O goes first

Qt Test App

— □ ×

Tic-Tac-Toe

O's turn

Back

Reset Board

